GATE-ALL BRANCH



Engineering Mathematics

PROBABILITY AND STATISTICS

Lecture No.- 03



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Topics to be Covered



Topic Problems based on fundamental probability.

$$P(E) = \frac{n(E)}{n(S)}$$

Recap of Previous Lecture



Topic Concepts based on fundamental probability

Topic : Fundamentals of Probability

(1X1) square choose

#Q. 4 squares are chosen at random on the chessboard. Find the probability that they lie along the same diagonal line.

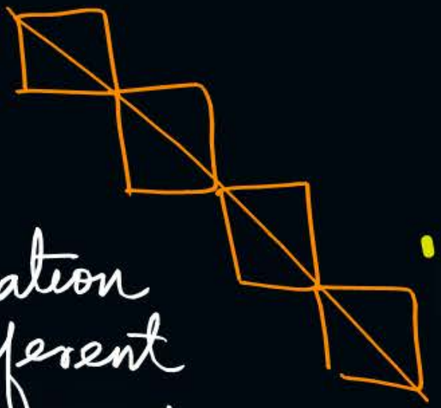
64 squares

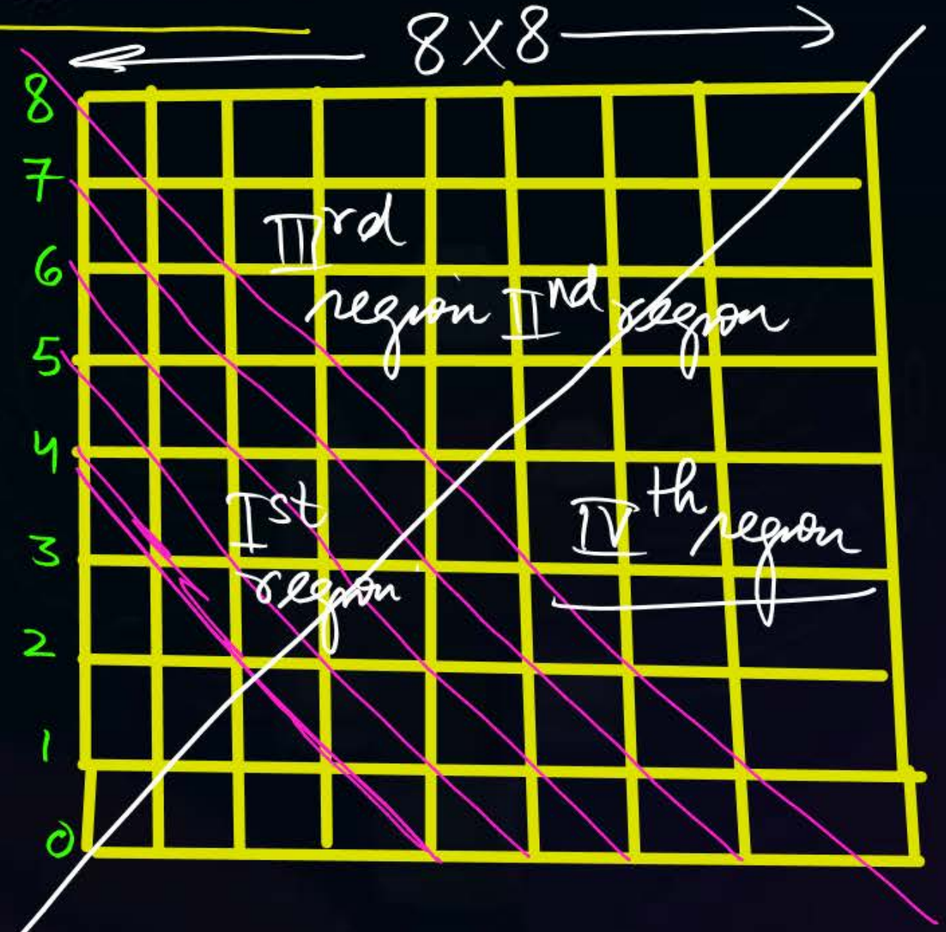
Choose
Combination
group
committee

A, B, C Three Letters

Combination
n Different Items r choose

$= {}^{n}C_{r} = \frac{\lfloor n}{\lfloor n-r \lfloor r}$





A, B, C 3 Different Items

2 Items Choose

✓ AB, BC, CA

X BA, CB, AC Order does Not matter

Arrangement
n Differents
r
$= {}^{n}P_{r}$
(Order Matter)

AB, BC, CA

Total choices $= {}^{3}C_{2}$

$= \frac{3!}{2!\,1!} = \frac{3\times 2\times 1}{2\times 1\times 1} = 3$

✓ n Different Items choose 'r' $\longrightarrow {}^{n}C_{r} = \frac{n!}{\lfloor n-r \rfloor r}$

n Different Items Arrange 'r' $\longrightarrow {}^{n}P_{r} = \frac{n!}{\lfloor n-r}$

n Different Items Taken all at a time $= n!$

I, II, III, IV

$$2\left[2\left[({}^{4}C_4 + {}^{5}C_4 + {}^{6}C_4 + {}^{7}C_4) + {}^{8}C_4\right]\right]$$

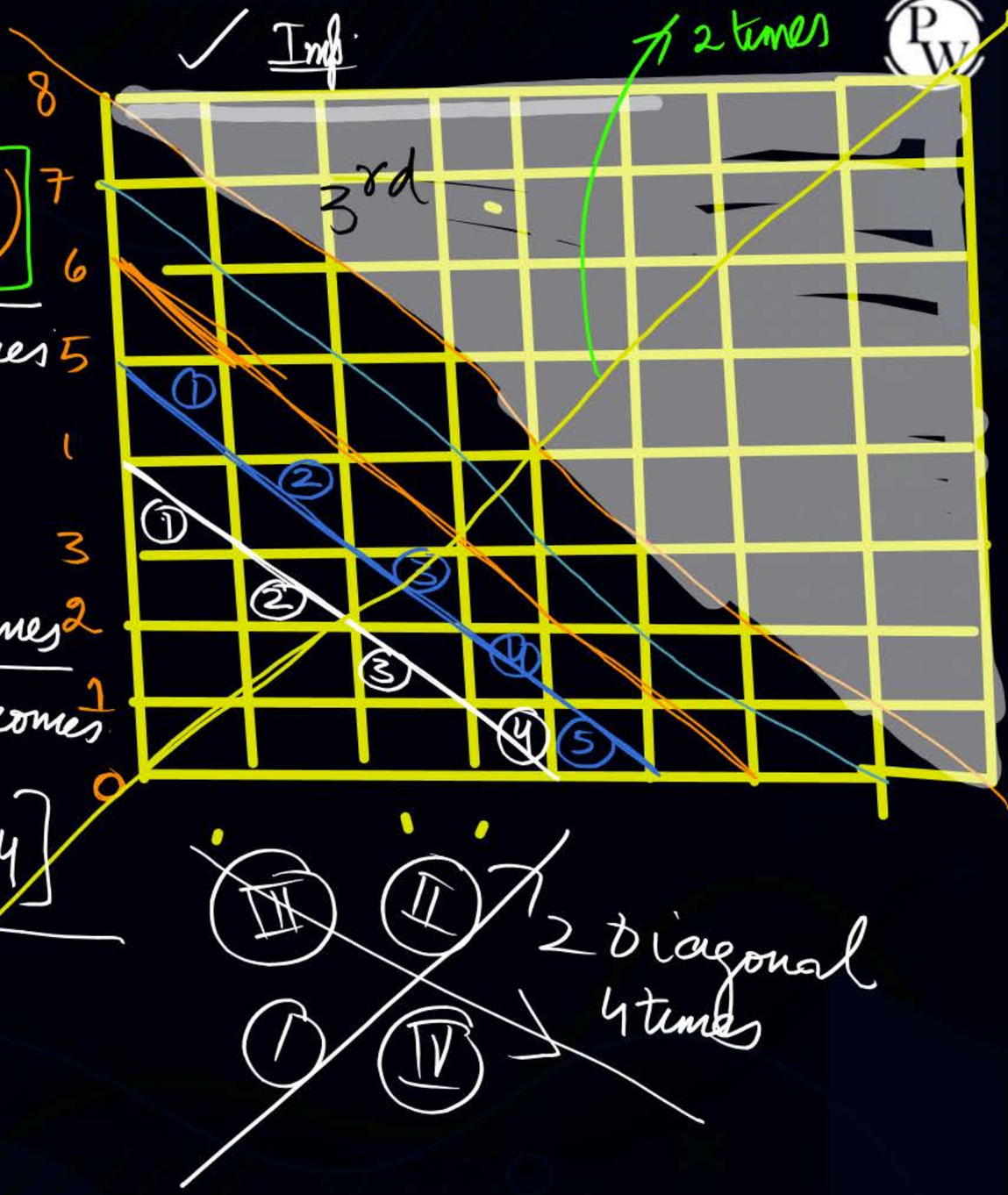
Favourable No. of outcomes

Total No. of outcomes $= {}^{64}C_4$

$$P(E) = \frac{\text{Favourable No. of outcomes}}{\text{Total No. of Possible outcomes}}$$

$$P(E) = \frac{2\left[2\left({}^{4}C_4 + {}^{5}C_4 + {}^{6}C_4 + {}^{7}C_4\right) + {}^{8}C_4\right]}{{}^{64}C_4}$$

Ans





Topic : Fundamentals of Probability

#Q. 4 squares are chosen at random on a chessboard. Find the probability that they lie along the same diagonal line adjacent to each other.

Topic : Fundamentals of Probability

#Q. 2 squares are chosen on a chessboard. Find the probability that they have a side in common.

$$P(E) = \frac{\text{No. of Favourable outcomes}}{\text{Total No. of Possible outcomes}}$$



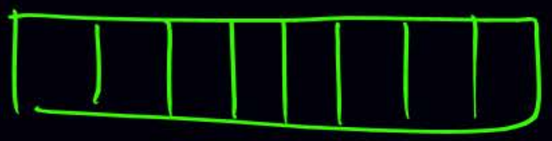
(1,2), (2,3), (3,4), (4,5), (5,6), (6,7), (7,8)

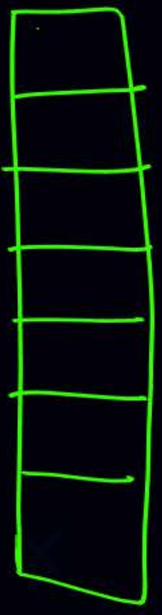
① ② ③ ④ ⑤ ⑥ ⑦

$= 7 \times 8 =$ Horizontal

$= 7 \times 8 =$ Vertical

$$\text{Prob.}(E) = \frac{7 \times 8 + 7 \times 8}{64C_2}$$ Ans





7 side common



common



$64C_2 = n(S)$



Topic : Fundamentals of Probability

$7 = H-T \; n=10$
$6 = H-T \; n=6$ (GATE)

$n = 3|4|5|6 \ldots$

#Q. A fair coin is tossed n times, the probability that the difference between the number of heads and tail is (n-3) is _____.

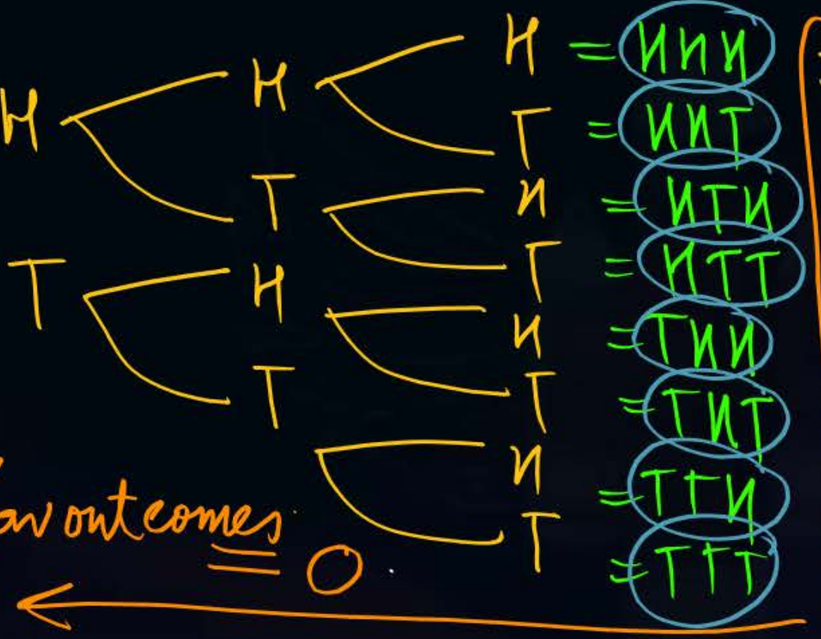
A Fair coin Is Tossed 3 times the Prob That The Diff $H-T = n-3$
$= 3-3$
$H-T = 0$



Total sAmple Pts

$P(E) = \frac{n(E)}{n(S)} = \frac{0}{8}$

$P(H-T=0) = 0$



3H, 0T = 3-0=3
2H, 1T = 2-1=1
2H, 1T = 2-1=1
1H, 2T = 1-2=-1
1T, 2H = 1-2=-1
2T, 1H = 2-1=1
2T, 1H = 2-1=1
3T = 3-0=3

for n times :

HEAD	TAIL	=(n-3) H-T	
n	0	n-0	n
n-1	1	n-1-1	n-2
n-2	2	n-2-2	n-4
n-3	3	n-3-3	n-6
⋮	⋮	⋮	
0	n		

$n-3$
$= 4-3$
$= 1$

Nowhere represent $(n-3)$

Single coin = 2 = 2
Two = 4 = 2^2
Three = 8 = 2^3

$\boxed{2^n}$

Total outcomes = ① ② ③ ④ ⑤ ⑥
$2 \times 2 \times 2 \times 2 \times 2 \times 2$ ... n times
$= 2^n$

favourable outcomes = 0

$$P(E) = \frac{0}{2^n} = 0$$



Topic : Fundamentals of Probability

#Q. If P and Q are chosen randomly from the set {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} with replacement. Determine the probability that the roots of the equation $x^2 + px + q = 0$ are real.

P ⟶ {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

q ⟶ {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

P, q

1, 2, 3, 4, 5

6, 7, 8, 9, 10

$x^2+px+q=0$

If Roots Are real $(p^2-4q\geq 0)$

$P[p^2-4q\geq 0]$

$P \longrightarrow$ 10 choices
$q \longrightarrow$ 10 choices with replacement

Total choice $= 10\times 10$
$= 100$

P 1 2 3 4 5 6 7 8 9 10

q 1 2 3 4 5 6 7 8 9 10

$ax^2+bx+c=0$

Roots Are real

$\boxed{b^2-4ac\geq 0}$

Roots Are real.



$p^2 = 1, 4, 9, 16, 25, 36, 49, 64, 81, 100$

$4q = 4, 8, 12, 16, 20, 24, 28, 32, 36, 40$

Fav No of choice $= p^2 - 4q \geq 0$

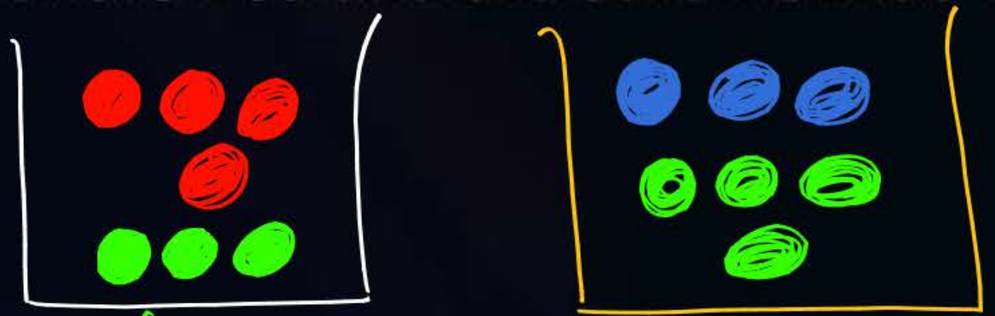
$\Rightarrow 0+1+2+4+6+9+10+10+10+10 = 62$

$$\text{Prob. } P(p^2 - 4q \geq 0) = \frac{62}{100}$$

$= P(p^2 - 4q \leq 0) =$ Do yourself

Topic : Fundamentals of Probability

#Q. There are two containers, with one containing 4 Red and 3 Green balls and the other containing 3 Blue balls and 4 Green balls. One ball is drawn at random from each container. The probability that one of the balls is Red and the other is Blue will be ____.



P(Red) × P(blue)

$\frac{4}{7} \times \frac{3}{7} = \frac{12}{49}$

Red blue



Topic : Fundamentals of Probability

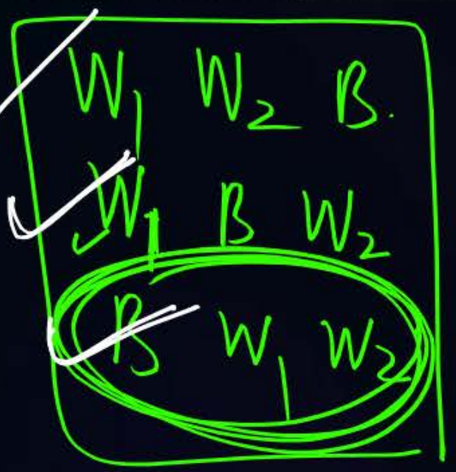
H.W

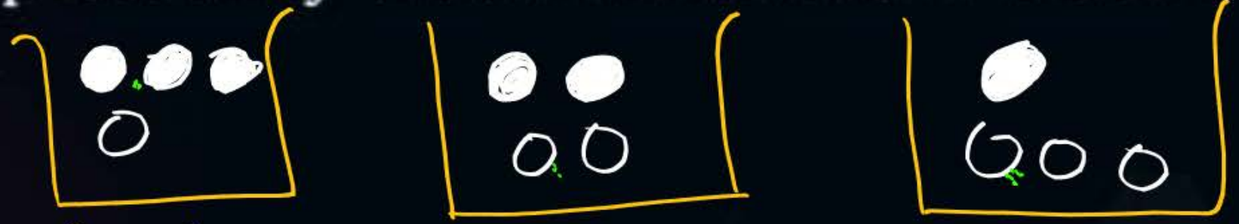
#Q. An urn contains 5 red and 7 green balls. A ball is drawn at random and its colour is noted. The ball is placed back into the urn along with another ball of the same colour. The probability of getting a red ball in the next draw is ____.

Topic : Fundamentals of Probability

13/32

#Q. If from each of the three boxes containing 3 white and 1 black balls, 2 white and 2 black balls, 1 white and 3 black balls, one ball is drawn at random, then the probability that 2 white and 1 black ball will be drawn is _____.





$W_1\ W_2\ B = \frac{3}{4} \times \frac{2}{4} \times \frac{3}{4} = \frac{18}{64}$

$W_1\ B\ W_2 = \frac{3}{4} \times \frac{2}{4} \times \frac{1}{4} = \frac{6}{64}$ +

$B\ W_1\ W_2 = \frac{1}{4} \times \frac{2}{4} \times \frac{1}{4} = \frac{2}{64}$ +

2W : 1B

$B\ W_2\ W_3$.

$= \frac{18}{64} + \frac{6}{64} + \frac{2}{64} = \frac{26}{64} = \frac{13}{32}$

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THANK - YOU